

The Mole Concept

Answer key with worked solutions

A. Understanding the mole

1. What is a mole?

A mole is the amount of substance that contains 6.022×10^{23} particles (atoms, molecules, ions or formula units).

2. Same number of atoms?

Yes. Both are one mole, so both contain 6.022×10^{23} atoms. The masses differ because a magnesium atom is heavier than a carbon atom.

3. What is wrong?

Oxygen gas is made of O_2 molecules. One mole contains 6.022×10^{23} *molecules*, which is $2 \times 6.022 \times 10^{23} = 1.204 \times 10^{24}$ oxygen *atoms*.

B. Molar mass

4. Molar masses.

(a) CH_4 : $12.01 + 4(1.008) = 16.04 \text{ g/mol}$

(b) NH_3 : $14.01 + 3(1.008) = 17.03 \text{ g/mol}$

(c) $CaCO_3$: $40.08 + 12.01 + 3(16.00) = 100.09 \text{ g/mol}$

5. Molar masses.

(a) $C_6H_{12}O_6$: $6(12.01) + 12(1.008) + 6(16.00) = 72.06 + 12.096 + 96.00 = 180.16 \text{ g/mol}$

(b) $MgCl_2$: $24.31 + 2(35.45) = 95.21 \text{ g/mol}$

C. Converting between mass, moles and particles

6. Moles in 25.0 g of CH₄.

$$n = \frac{m}{M} = \frac{25.0 \text{ g}}{16.04 \text{ g/mol}} = \mathbf{1.56 \text{ mol}}$$

7. Mass of 0.150 mol of CaCO₃.

$$m = n \times M = 0.150 \text{ mol} \times 100.09 \text{ g/mol} = \mathbf{15.0 \text{ g}}$$

8. Molecules in 0.500 mol of NH₃.

$$N = 0.500 \text{ mol} \times 6.022 \times 10^{23} \text{ mol}^{-1} = \mathbf{3.01 \times 10^{23} \text{ molecules}}$$

9. Moles in 1.50×10^{24} molecules of water.

$$n = \frac{1.50 \times 10^{24}}{6.022 \times 10^{23} \text{ mol}^{-1}} = \mathbf{2.49 \text{ mol}}$$

10. Mass of 3.01×10^{23} sodium atoms.

$$n = \frac{3.01 \times 10^{23}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 0.500 \text{ mol}$$
$$m = 0.500 \text{ mol} \times 22.99 \text{ g/mol} = \mathbf{11.5 \text{ g}}$$

D. Counting atoms inside compounds

11. Moles of H atoms in 2.00 mol of glucose.

$$\text{Each C}_6\text{H}_{12}\text{O}_6 \text{ contains 12 H atoms: } 2.00 \text{ mol} \times 12 = \mathbf{24.0 \text{ mol of H atoms}}$$

12. Oxygen atoms in 10.0 g of CO₂.

$$M(\text{CO}_2) = 12.01 + 2(16.00) = 44.01 \text{ g/mol}$$
$$n(\text{CO}_2) = \frac{10.0}{44.01} = 0.2272 \text{ mol}$$
$$\text{O atoms} = 2 \times 0.2272 \text{ mol} \times 6.022 \times 10^{23} \text{ mol}^{-1} = \mathbf{2.74 \times 10^{23} \text{ atoms}}$$

13. Mass of chlorine in 1.00 mol of MgCl₂.

$$1.00 \text{ mol of MgCl}_2 \text{ contains 2.00 mol of Cl: } 2.00 \text{ mol} \times 35.45 \text{ g/mol} = \mathbf{70.90 \text{ g}}$$

E. Challenge

14. More molecules: 10.0 g of H₂O or 10.0 g of CO₂?

$$n(\text{H}_2\text{O}) = \frac{10.0}{18.02} = 0.555 \text{ mol} \quad n(\text{CO}_2) = \frac{10.0}{44.01} = 0.227 \text{ mol}$$

Water contains more molecules (3.34×10^{23} compared with 1.37×10^{23}). Water molecules are lighter, so the same mass contains more of them.

15. Identify the element.

$$M = \frac{m}{n} = \frac{5.00 \text{ g}}{0.125 \text{ mol}} = 40.0 \text{ g/mol. This matches calcium (Ca = 40.08).}$$